

Notes of HDP¹

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Outline

Introduction

Concentration inequalities

Variance Bounds

- Tensorization property of the Variance
- Bounded Difference Inequalities, Variance
- Poincaré Inequalities

Subgaussian Concentration

- Subgaussianity
- Bounded Difference Inequalities, Subgaussian weak version
- Tensorization property of the Entropy
- Bounded Difference Inequalities, Subgaussian strong version
- Log-Sobolev Inequalities

Sub-Exponential Concentration

Suprema and Chaining

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Intro

Informal version of

- ▶ Concentration
- ▶ Suprema

Preliminaries and Notations

- ▶ $a \lesssim b$: $a \leq Cb$ where C is some absolute constant.

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Intro - Concentration inequalities

(1) Variance version of concentration

$$\mathrm{Var}(f) \lesssim \sum_{i=1}^n \mathbb{E} [(D_i f)^2]$$

the variance of a function f of independent (or weakly dependent) random variables is small if the “gradient” of f is small.

(2) Sub-Gaussianity, a sharper control on the distribution of the fluctuations

$$\|f - \mathbb{E}f\|_{\psi_2}^2 \lesssim \left\| \sum_{i=1}^n D_i f \right\|_{\infty}^2 \quad \text{by simple martingale}$$

$$\|f - \mathbb{E}f\|_{\psi_2}^2 \lesssim \left\| \sum_{i=1}^n D_i f \right\|_{\infty}^2 \quad \text{by entropy method}$$



(3)

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Tensorization property of the Variance

Efron-Stein

- relationship between the variance of the overall function and the sum of the individual expected variance[] contributions

[Efron-Stein] Let $X_1, \dots, X_n$ be independent random variables, define

$$\text{Var}_i f(x_1, \dots, x_n) := \text{Var}[f(x_1, \dots, x_{i-1}, \mathbf{x}_i, x_{i+1}, \dots, x_n)] = \text{Var}(f \mid X_{-i})$$

Then

$$\text{Var}[f(X_1, \dots, X_n)] \leq \sum_{i=1}^n \mathbb{E} [\text{Var}_i f(X_1, \dots, X_n)]$$

Bounded Difference Inequalities

Variance version

- relationship between variance of overall function and the sum of Difference $D_i f$ with respect to individual variable

[Bounded Difference Inequalities] Let $X_1, \dots, X_n$ be independent random variables, define

$$D_i f(x_1, \dots, x_n) = \sup_{x_i} f - \inf_{x_i} f$$

$$D_i^- f(x_1, \dots, x_n) = f - \inf_{x_i} f$$

Then

$$\text{Var}(f) \leq \frac{1}{4} \sum_{i=1}^n \mathbb{E} [(D_i f)^2]$$

$$\text{Var}(f) \leq \frac{1}{4} \sum_{i=1}^n \mathbb{E} [(D_i^- f)^2]$$

Example:

$D_i f$, a 'discrete derivative' of the function with respect to the i -th variable

Poincaré Inequalities

- ▶ from 'variance/difference' to 'derivative'

$$\text{Var}(f) \lesssim \mathbb{E} [\|\text{gradient}(f)\|^2]$$

[Convex Poincaré Inequalities] Let $\{X_i\}_{i=1}^n$ be independent random variables, $X_i \in [a, b]$, $f : \mathbb{R}^n \rightarrow \mathbb{R}$ convex and differentiable. Then

$$\text{Var}(f) \leq (b - a)^2 \cdot \mathbb{E} \|\nabla f(X_1, \dots, X_n)\|_2^2$$

proof: first relate 'difference' with 'derivative' with respect to certain assumption on X and f , then use efron-stein/bounded difference inequalities

Poincaré Inequalities

General version

Poincaré Inequalities

Application to Sampling

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Subgaussian Concentration

- ▶ Subgaussianity: another way to describe concentration
- ▶ Entropy Method
- ▶ Log-Sobolev Inequalities

Tail probability bounded by 'Moments'

Chebyshev inequality, Chernoff bound

Let X be a random variable. Then, for any $t > 0$, we have

- ▶ -[Chebyshev inequality]

$$\mathbb{P}\{|X - \mathbb{E}X| \geq t\} \leq \frac{\mathbb{E}|X - \mathbb{E}X|^p}{t^p}$$

- ▶ -[Chernoff bound, exponential moment method] Define the log-moment generating function ψ of a random variable X and its Legendre dual ψ^* as

$$\psi(\lambda) := \log \mathbb{E}e^{\lambda(X - \mathbb{E}X)}, \quad \psi^*(t) = \sup_{\lambda \geq 0} \{\lambda t - \psi(\lambda)\}.$$

Then

$$\mathbb{P}\{X - \mathbb{E}X \geq t\} \leq e^{-\psi^*(t)} \quad \text{for all } t \geq 0.$$

bounding tail probability $\rightarrow$ bounding the MGF.

proof: by Markov Inequality on nonnegative increasing function of $X - \mathbb{E}X$

Subgaussian Variable

[Subgaussian properties] Let X be a random variable. The follows are equivalent, with $K_i > 0$ differing by at most an absolute constant.

(i) (Tail bound) There exists $K_1 > 0$ such that

$$\mathbb{P}\{|X| \geq t\} \leq 2 \exp(-t^2/K_1^2) \quad \text{for all } t \geq 0.$$

(ii) (Moment growth bound) There exists $K_2 > 0$ such that

$$\frac{\|X\|_{L^p}}{\sqrt{p}} = \frac{(\mathbb{E}|X|^p)^{1/p}}{\sqrt{p}} \leq K_2 \quad \text{for all } p \geq 1.$$

[Subgaussian norm] $\|X\|_{\psi_2} = \sup_{p \geq 1} \left\{ \frac{(\mathbb{E}|X|^p)^{1/p}}{\sqrt{p}} \right\}$, the smallest K_2 .

(iii) (Finite MGF of X^2) There exists $K_3 > 0$ such that

$$\mathbb{E} \exp(X^2/K_3^2) \leq 2.$$

Moreover, if $\mathbb{E}X = 0$, (i)–(iii) are equivalent to the following one:

(iv) (MGF bound) There exists $K_4 > 0$ such that

$$\mathbb{E} \exp(\lambda X) \leq \exp(\lambda^2 K_4^2) \quad \text{for all } \lambda \in \mathbb{R}.$$

[Centering] Any subgaussian random variable X satisfies

$$\|X - \mathbb{E}X\|_{\psi_2} \leq C \|X\|_{\psi_2}.$$

Subgaussianity

a centered version

[Subgaussianity] A random variable X is σ^2 -subgaussian if

$$\mathbb{E}e^{\lambda(X-\mathbb{E}X)} \leq e^{\lambda^2\sigma^2/2}, \quad \forall \lambda \in \mathbb{R}$$

If X is σ^2 -subgaussian, then

$$\mathbb{P}\{X - \mathbb{E}X \geq t\} \leq e^{-t^2/2\sigma^2}, \quad \mathbb{P}\{X - \mathbb{E}X \leq -t\} \leq e^{-t^2/2\sigma^2}.$$

This is also equivalent to

$$\|X - \mathbb{E}X\|_{\psi_2}^2 \lesssim \sigma^2.$$

σ^2 is called the variance proxy because it shows like variance, the sum of n independent σ^2 -subg variable is $\frac{1}{n}\sigma^2$ -subg.

Subgaussianity

What r.v.'s are subgaussian?

- ▶ Gaussian
- ▶ A bounded r.v. [Hoeffding] If $a \leq X \leq b$ a.s. for constants a & b , then X is $\frac{(b-a)^2}{4}$ -subg.

Subgaussianity of a function

Bounded Difference Inequalities

Subgaussianity is another description of concentration, and now we discuss the subgaussianity of function $f(X_1, \dots, X_n)$.

[McDiarmid] If $\{X_i\}_{i=1}^n$ are indep, $\|D_i f\|_\infty < \infty$ for all i , then $f(X_1, \dots, X_n)$ is subgaussian with variance proxy $\frac{1}{4} \sum_{i=1}^n \|D_i f\|_\infty^2$

$$\mathbb{P}\{f - \mathbb{E}f \geq t\} \leq e^{-2t^2 / \sum_{i=1}^n \|D_i f\|_\infty^2}$$

$$\|f - \mathbb{E}f\|_{\psi_2}^2 \lesssim \sum_{i=1}^n \|D_i f\|_\infty^2.$$

Actually subgaussian property does not tensorize naturally, and the martingale method can only partially address this issue. Next, we explore the entropy method to derive a better bound.

Entropy and Subgaussian

[Entropy] The entropy of a nonnegative random variable Z is

$$\text{Ent}[Z] := \mathbb{E}[Z \log Z] - \mathbb{E}[Z] \log \mathbb{E}[Z].$$

A sufficient condition for subgaussian in terms of entropy:

[Herbst] If $\text{Ent}(e^{\lambda X}) \leq \frac{\lambda^2 \sigma^2}{2} \mathbb{E} e^{\lambda X}$, $\forall \lambda$, then X is σ^2 -subg.

Tensorization property of the Entropy

Let $X_1, \dots, X_n$ be independent random variables, $f \geq 0$, define

$$g_i(x_1, \dots, x_n) = \text{Ent} [f(x_1, \dots, x_{i-1}, X_i, x_{i+1}, \dots, x_n)]$$

then

$$\text{Ent} [f(X_1, \dots, X_n)] \leq \sum_{i=1}^n \mathbb{E} [g_i(X_1, \dots, X_n)]$$

Bounded Difference Inequalities

a stronger version derived by entropy method

If $\{X_i\}_{i=1}^n$ indep. $\|D_i f\|_\infty < \infty$ for all i , then

$$\|f - \mathbb{E}f\|_{\psi_2}^2 \lesssim \left\| \sum_{i=1}^n |D_i f|^2 \right\|_\infty$$

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Proof of Efron-Stein

Proof of Hoeffding

Proof of McDiarmid

references